



1T □□□□□□□□□□□□□□-□□□□□□

Investigation of Current-Phase Relationship in 1T-Phase Platinum Ditelluride Josephson Junctions under In-Plane Magnetic Field

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□□ (Introduction)

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1. □□

- □□□□
- □□□□□□□□ (1T-PtTe₂)

2. □□□□

- □□□□□□□□-□□□□ (CPR)
- □□□□□□□□ (JDE)

3. □□□□

- □□□□□□
- □□□□□□

4. □□□□□

- □□□□□□
- □□-□□□□ (CPR) □□
- □□□□□□□□

5. □□

□□□□ (Experimental Methods)

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1. □□□□: □□□□ (□□□□□□□□□□□□)□
2. □□□□: □ PtTe_2 □□□□□ SiO_2/Si □□□□
3. □□□□□ (EBL):
 - □□ PMMA □□□
 - □□□□□□□ SQUID □□□□□
4. □□□□□:
 - □□□□□□: □ (Al) □ □ (Nb)□
 - □□ (*In-situ*) □□□□□□□□□□□□□□□□
 - □□□□□□□□ (Lift-off)□
5. □□□□: □□□□□□□□□□□□ (Asymmetric dc-SQUID) □□□□□□□□□ CPR□

□□□□□□

- □□□□□: □□□□□ (DR)
- □□: $\sim 40 \text{ mK}$ □

□□□□□□□□ (Basic Characterization)

I-V □□□□ (□□□□)

- □□ IV □□:
 - □□□□□□□□ ($V = 0$)□
 - □□□□ (I_C) □□□□□□ (I_r)□
 - □□□□ (□□□□□□)□

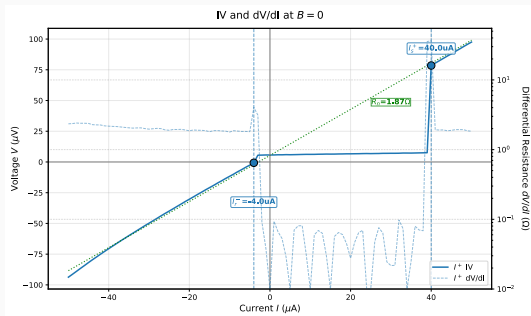


Figure 3: □□□□ I-V □□

- □□□□:
 - □□□□□□□□□□□□□□□□
 - □□□□□□□□□□ (R_N)□
 - □□□□□□□□ □□□□ (Excess Current) $\rightarrow$ □□□□□□□□□□

□□□□□-□□□□ (Current-Phase Relation)

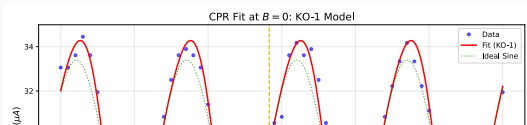
CPR □□□□

- **SQUID** □□:
 - (I_{ref}) // □□□ (I_{test})□
 - $I_c(\Phi_{ext}) \approx I_{ref} + I_{test}(\phi)$.
 - $I_{ref} \gg I_{test}$ □ $I_{ref}(\phi)$ □□□□□□
- :

$$I_s(\phi) = I_1 \sin(\phi) + I_2 \sin(2\phi + \delta)$$

- I_1 : □□□□ (□□□□)□
- I_2 : □□□□□ (4π -□□ / ABS)□

□□ CPR □□□□□

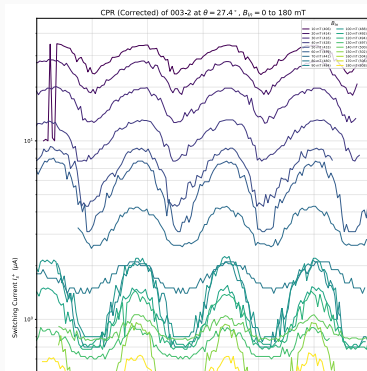


- :
 - **CPR**: □□□□□□□□□□
 - I_2/I_1 □□:
□□□□□□□□□□□□□□

□□□□□□□□ (In-Plane Magnetic Field Effect)

□□□□□ CPR

- :
 - $B_{\parallel}$ □□□ CPR □□□□
 - □□□□ (ϕ_0)□
 - (JDE) □□□□



□□ (Conclusion)

□□

1. □□□□:

- □□□□□ 1T-PtTe₂ □□□□□□□ SQUID □□□

2. □□ **CPR**:

- □□□□□□ □□□□ (I_2) □□□
- □□□□□□□□□□□□□□

3. □□□□:

- □□□□□□□□□ CPR □□□ □□□□□□□□□□
- □□□□□ □□□□□□□□ □ □□□□ □□□□□□□

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- □□□□: □□□□□□□□□□□□□□□□□□
- □□□□: □□□□□□□□ (Weyl) / □□□□ (Dirac) □□□□
- □□□□: □□□□□□□□□□□□□□ (□□□□□□□□□□□□□□)□

